

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Mock Interview

Course Code:	ITA05	Course Title:	Computer Vision
CO Assessed:	CO5 – Naturalization (independent application using OpenCV)P5	Assessment:	Assessment Tool 3 – Mock Interview
Weightage:	25% of CIA	Total Marks:	100
CO4 Statement:	Develop computer vision applications using OpenCV for performing recognition and analysis on images and videos. (BTL6)		
Student Name:	KAVIYA V	Reg. No.:	192421022
Section / Batch:	Third Year/ B.Tech IT	Date:	26/08/2026

Code of Conduct

I _Kaviya v / 192421022_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _Kaviya V_____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge	
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts	
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated	
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively	
Confidence	10	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism	
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately	

QUESTION:

Face & Expression Recognition - 12. A face detection system produces multiple false positives. Analyze the issue and propose improvements, explaining how each change enhances accuracy.

ANSWERS:

12. Face & Expression Recognition – False Positive Analysis

Problem:

A face detection system produces multiple false positives, meaning it identifies non-face regions as faces. This reduces the reliability and accuracy of the system.

Analysis and Improvements:

1. Improve Image Preprocessing:

- Apply noise reduction, grayscale conversion, and suitable contrast enhancement.
- This makes facial features clearer and reduces incorrect detections caused by image noise.

2. Use a Better Face Detection Model:

- Replace a basic detector such as Haar Cascade with a more robust detector such as a modern CNN/DNN-based detector.
- Modern models can distinguish faces from background objects more effectively.

3. Adjust Detection Threshold:

- Increase the confidence threshold for accepting a detected face.
- Low-confidence detections are rejected, reducing false positives.

4. Use Non-Maximum Suppression (NMS):

- When several overlapping boxes identify the same region, NMS keeps the strongest detection and removes redundant boxes.
- This produces cleaner detection results.

5. Improve Training Data:

- Train or fine-tune the model using more diverse images containing different lighting conditions, poses, backgrounds, and face sizes.
- This improves generalization and reduces incorrect detections.

6. Restrict the Region of Interest (ROI):

- Detect faces only in relevant areas of the image instead of searching the entire frame.
- This reduces the possibility of background regions being classified as faces.

Conclusion:

The false positives can be reduced by improving preprocessing, using a more robust detection model, increasing the confidence threshold, applying NMS, using diverse training data, and restricting detection to suitable regions. These changes reduce background misclassification and improve the accuracy and reliability of face detection.